OBJECT ORIENTED PROGRAMMING USING JAVA

Units 1–5 | Part A – 1 Mark Multiple-Choice Questions with Options and Correct Answers
Based on the uploaded KPR Institute Java Question Bank (U25CSG11)

UNIT 1 – JAVA FUNDAMENTALS

Q1. Which programming paradigm organizes a program into objects containing data and methods?

A. Procedure-Oriented Programming
B. Object-Oriented Programming
C. Functional Programming
D. Structured Programming
Correct Answer: B. Object-Oriented Programming

Q2. Which component executes Java bytecode?

A. JDK
B. JRE
C. JVM
D. Compiler
Correct Answer: C. JVM

Q3. Which is included in the JDK but not in the JRE?

A. JVM
B. Java API
C. Java compiler (javac)
D. Bytecode verifier
Correct Answer: C. Java compiler (javac)

Q4. Which method is the entry point of a Java application?

A. display()
B. execute()
C. start()
D. main()
Correct Answer: D. main()

Q5. Which statement correctly creates a Student object?

A. Student();
B. Student s;
C. Student s = new Student();
D. new Student s();
Correct Answer: C. Student s = new Student();

Q6. What is the output of System.out.println("Java");?

A. Java
B. java
C. Compilation error
D. No output
Correct Answer: A. Java

Q7. Which keyword is used to create an object in Java?

A. class
B. object
C. new
D. create
Correct Answer: C. new

Q8. Where are objects created using new stored?

A. Stack memory
B. Heap memory
C. Cache memory
D. Register memory
Correct Answer: B. Heap memory

Q9. Where are local variables generally stored?

A. Heap memory
B. Stack memory
C. ROM
D. Cache
Correct Answer: B. Stack memory

Q10. Which declaration represents an instance variable?

A. static int x;
B. int x;
C. void x(){}
D. None of these
Correct Answer: B. int x;

Q11. If static int x = 20, what is printed by System.out.println(x)?

A. 0
B. 20
C. Compilation error
D. Runtime error
Correct Answer: B. 20

Q12. How can a static method display() be invoked inside the same class?

A. obj.display();
B. display();
C. new display();
D. call display();
Correct Answer: B. display();

Q13. Which constructor is invoked by new Student(100)?

A. Default constructor
B. Copy constructor
C. Parameterized constructor
D. Destructor
Correct Answer: C. Parameterized constructor

Q14. Which constructor is supplied by the compiler if no constructor is declared?

A. Parameterized constructor
B. Copy constructor
C. Default constructor
D. Static constructor
Correct Answer: C. Default constructor

Q15. Which access modifier restricts access to the same class?

A. public
B. protected
C. private
D. default
Correct Answer: C. private

Q16. Hiding data with private members and providing getters/setters demonstrates:

A. Inheritance
B. Polymorphism
C. Encapsulation
D. Abstraction
Correct Answer: C. Encapsulation

Q17. Which class is commonly used for keyboard input?

A. System
B. Scanner
C. Input
D. ConsoleReader
Correct Answer: B. Scanner

Q18. Which statement correctly creates a Scanner for keyboard input?

A. Scanner sc = new Scanner();
B. Scanner sc = new Scanner(System.in);
C. Scanner sc = Scanner(System.in);
D. Scanner(System.in);
Correct Answer: B. Scanner sc = new Scanner(System.in);

Q19. Which statement displays the value of int x = 50?

A. Scanner.println(x);
B. System.in.println(x);
C. System.out.println(x);
D. print(x);
Correct Answer: C. System.out.println(x);

Q20. Identify the errors in public static void Main(...) and system.out.println(...).

A. Main should be main and system should be System
B. println should be print
C. args should be arg
D. No error
Correct Answer: A. Main should be main and system should be System

UNIT 2 – INHERITANCE AND POLYMORPHISM

Q1. Which keyword is used for inheritance in Java?

A. implement
B. extends
C. inherit
D. super
Correct Answer: B. extends

Q2. class B extends A demonstrates which inheritance type?

A. Multiple
B. Multilevel
C. Single
D. Hierarchical
Correct Answer: C. Single

Q3. Two display methods with int and double parameters demonstrate:

A. Method overloading
B. Method overriding
C. Inheritance
D. Abstraction
Correct Answer: A. Method overloading

Q4. What is required for method overriding?

A. Different method name
B. Same method signature
C. Different return type only
D. Static method
Correct Answer: B. Same method signature

Q5. Animal a = new Dog(); illustrates:

A. Encapsulation
B. Upcasting
C. Constructor chaining
D. Package creation
Correct Answer: B. Upcasting

Q6. Which Java feature supports multiple behavior inheritance?

A. Classes
B. Interfaces
C. Packages
D. Constructors
Correct Answer: B. Interfaces

Q7. Which keyword declares an abstract class?

A. interface
B. final
C. abstract
D. static
Correct Answer: C. abstract

Q8. Can an abstract class have a constructor?

A. Yes
B. No
C. Only if constructor is static
D. Only if constructor is final
Correct Answer: A. Yes

Q9. Which method does not have a body?

A. Static
B. Final
C. Abstract
D. Private
Correct Answer: C. Abstract

Q10. A class that does not implement all interface methods must be declared:

A. Abstract
B. Final
C. Static
D. Private
Correct Answer: A. Abstract

Q11. Which package contains Scanner?

A. java.io
B. java.util
C. java.lang
D. java.net
Correct Answer: B. java.util

Q12. Which package is imported automatically?

A. java.util
B. java.lang
C. java.io
D. java.awt
Correct Answer: B. java.lang

Q13. Which keyword connects a class to an interface?

A. extends
B. imports
C. implements
D. package
Correct Answer: C. implements

Q14. Animal a = new Dog(); a.sound(); demonstrates:

A. Overloading
B. Overriding
C. Constructors
D. Packages
Correct Answer: B. Overriding

Q15. Runtime polymorphism is achieved through:

A. Overriding
B. Overloading
C. Inheritance alone
D. Encapsulation
Correct Answer: A. Overriding

Q16. Which package contains File?

A. java.io
B. java.net
C. java.util
D. java.lang
Correct Answer: A. java.io

Q17. Which member is not inherited in Java?

A. Methods
B. Constructors
C. Variables
D. Protected members
Correct Answer: B. Constructors

Q18. Which access modifier permits access between classes in the same package without public visibility?

A. public
B. protected
C. private
D. default
Correct Answer: D. default

Q19. Which keyword defines a package?

A. import
B. package
C. extends
D. implements
Correct Answer: B. package

Q20. Interface variables are implicitly:

A. public static final
B. private
C. protected
D. static only
Correct Answer: A. public static final

UNIT 3 – EXCEPTION HANDLING AND MULTITHREADING

Q1. Which keyword manually throws an exception?

A. throw
B. throws
C. catch
D. except
Correct Answer: A. throw

Q2. If division by zero is inside try and ArithmeticException is caught, what happens?

A. Compiler error
B. Program crashes
C. Catch executes and program continues
D. Infinite loop
Correct Answer: C. Catch executes and program continues

Q3. What is the superclass of all Java exception classes?

A. Exception
B. Throwable
C. Error
D. RuntimeException
Correct Answer: B. Throwable

Q4. An invalid array index occurs, but only NullPointerException is caught. What happens?

A. Caught silently
B. Uncaught exception terminates the program
C. Compiler error
D. Array auto-resizes
Correct Answer: B. Uncaught exception terminates the program

Q5. If both try and finally return values, which return takes precedence?

A. try value
B. finally value
C. Both
D. Compiler error
Correct Answer: B. finally value

Q6. ArithmeticException is:

A. Checked
B. Unchecked
C. Error
D. User-defined
Correct Answer: B. Unchecked

Q7. An unhandled checked IOException causes:

A. Normal execution
B. Compile-time error
C. Runtime warning
D. Silent ignore
Correct Answer: B. Compile-time error

Q8. What happens if an exception is never caught?

A. Program continues normally
B. Program terminates abnormally
C. Compiler error
D. Nothing
Correct Answer: B. Program terminates abnormally

Q9. catch(Exception) before catch(ArithmeticException) causes:

A. Runs normally
B. Unreachable catch compile-time error
C. Only arithmetic catch executes
D. Both execute
Correct Answer: B. Unreachable catch compile-time error

Q10. Multiple catch blocks should be ordered:

A. Any order
B. Superclass first
C. Subclass before superclass
D. Random order
Correct Answer: C. Subclass before superclass

Q11. Calling t.run() instead of t.start() means:

A. New thread starts
B. run executes on the current thread
C. Compiler error
D. Program hangs
Correct Answer: B. run executes on the current thread

Q12. Which method actually starts a new thread?

A. run()
B. start()
C. init()
D. execute()
Correct Answer: B. start()

Q13. Do thread priorities guarantee execution order?

A. Priority 10 always first
B. No; scheduler order is not guaranteed
C. Priority 1 always first
D. Start order is guaranteed
Correct Answer: B. No; scheduler order is not guaranteed

Q14. Which method pauses a thread for a specified time?

A. wait()
B. sleep()
C. join()
D. yield()
Correct Answer: B. sleep()

Q15. Which keyword provides mutual exclusion?

A. synchronized
B. volatile
C. static
D. transient
Correct Answer: A. synchronized

Q16. Calling t.join() from main makes:

A. Main wait until t finishes
B. t wait for main
C. No effect
D. t terminate
Correct Answer: A. Main wait until t finishes

Q17. Unsynchronized increments of a shared counter can cause:

A. Compiler error
B. Race condition
C. Deadlock always
D. StackOverflowError
Correct Answer: B. Race condition

Q18. Two threads waiting indefinitely for each other's locks is:

A. Starvation
B. Deadlock
C. Race condition
D. Livelock
Correct Answer: B. Deadlock

Q19. If wait() is called and no thread calls notify(), the waiting thread:

A. Resumes after sleep
B. Waits indefinitely
C. Throws immediately
D. Gets compile error
Correct Answer: B. Waits indefinitely

Q20. Calling run() directly instead of start() results in:

A. A new thread
B. Code running in the calling thread
C. Compiler error
D. Runtime exception
Correct Answer: B. Code running in the calling thread

UNIT 4 – COLLECTIONS FRAMEWORK AND GENERICS

Q1. Which package contains the Collections Framework?

A. java.util
B. java.io
C. java.lang
D. java.net
Correct Answer: A. java.util

Q2. Which feature provides compile-time type safety for ArrayList<String>?

A. Serialization
B. Inheritance
C. Generics
D. Reflection
Correct Answer: C. Generics

Q3. Which wrapper class corresponds to char?

A. Character
B. Char
C. String
D. Object
Correct Answer: A. Character

Q4. Which queue is suitable when highest-priority patients are served first?

A. LinkedList
B. ArrayDeque
C. PriorityQueue
D. Stack
Correct Answer: C. PriorityQueue

Q5. Which interface traverses collection elements?

A. Comparable
B. Iterator
C. Comparator
D. Enumeration
Correct Answer: B. Iterator

Q6. Which Iterator method should be checked before next()?

A. next()
B. hasNext()
C. contains()
D. iterator()
Correct Answer: B. hasNext()

Q7. Which interface provides natural ordering?

A. Comparator
B. Comparable
C. Cloneable
D. Serializable
Correct Answer: B. Comparable

Q8. Which collection stores unique employee IDs?

A. ArrayList
B. LinkedList
C. HashSet
D. Vector
Correct Answer: C. HashSet

Q9. Which collection preserves insertion order and prevents duplicates?

A. TreeSet
B. HashSet
C. LinkedHashSet
D. PriorityQueue
Correct Answer: C. LinkedHashSet

Q10. Which implementation stores elements in ascending sorted order?

A. HashSet
B. TreeSet
C. ArrayList
D. LinkedList
Correct Answer: B. TreeSet

Q11. Which implementation uses a Red-Black Tree internally?

A. HashMap
B. TreeMap
C. LinkedList
D. ArrayList
Correct Answer: B. TreeMap

Q12. Which interface provides key-value mapping?

A. List
B. Queue
C. Map
D. Stack
Correct Answer: C. Map

Q13. Which map implementation permits one null key?

A. TreeMap
B. HashMap
C. Hashtable
D. TreeSet
Correct Answer: B. HashMap

Q14. Which declaration creates a generic ArrayList of Strings?

A. ArrayList list=new ArrayList();
B. ArrayList<String> list=new ArrayList<>();
C. ArrayList<> list;
D. List=int;
Correct Answer: B. ArrayList<String> list=new ArrayList<>();

Q15. Which statement demonstrates autoboxing?

A. Integer i = Integer.valueOf(10);
B. Integer i = 10;
C. int i = Integer.parseInt("10");
D. Integer i = new Integer(10);
Correct Answer: B. Integer i = 10;

Q16. Which statement demonstrates unboxing?

A. Integer obj = 20;
B. int x = obj;
C. Integer.valueOf(20);
D. Integer.parseInt("20");
Correct Answer: B. int x = obj;

Q17. Which wrapper class corresponds to primitive float?

A. Double
B. Float
C. Decimal
D. Number
Correct Answer: B. Float

Q18. For list containing A then B, what does list.get(1) return?

A. A
B. B
C. null
D. Exception
Correct Answer: B. B

Q19. After map.put(1,"Java") then map.put(1,"Python"), what is map.size()?

A. 0
B. 1
C. 2
D. Compilation error
Correct Answer: B. 1

Q20. After adding 30, 20, 40 to a TreeSet, what is printed?

A. [30,20,40]
B. [20,30,40]
C. [40,30,20]
D. Random order
Correct Answer: B. [20,30,40]

UNIT 5 – JAVA FILE I/O AND SERIALIZATION

Q1. If data.txt does not exist, what do exists() calls print before and after FileWriter creates it?

A. false, false
B. true, true
C. false, true
D. Compilation error
Correct Answer: C. false, true

Q2. Which class writes int, double and boolean values to a binary file?

A. FileWriter
B. PrintWriter
C. DataOutputStream
D. BufferedWriter
Correct Answer: C. DataOutputStream

Q3. Scanner.nextInt() receives the text "twenty". Which exception is thrown?

A. NumberFormatException
B. InputMismatchException
C. IllegalStateException
D. NoSuchElementException
Correct Answer: B. InputMismatchException

Q4. What is efficient for reading a large text file line by line?

A. FileReader alone
B. BufferedReader wrapping FileReader
C. FileInputStream alone
D. Scanner on System.in
Correct Answer: B. BufferedReader wrapping FileReader

Q5. After deserializing an object with transient double salary = 50000, salary becomes:

A. 50000.0
B. 0.0
C. null
D. Compilation error
Correct Answer: B. 0.0

Q6. Writing the same object reference twice to the same ObjectOutputStream generally writes:

A. Two independent full copies
B. Second call throws an exception
C. A back-reference handle
D. Second call is ignored
Correct Answer: C. A back-reference handle

Q7. When can System.console() return null?

A. When run from an IDE
B. When run from a command-line terminal
C. When file is missing
D. When Scanner is also used
Correct Answer: A. When run from an IDE

Q8. What is the return type of Files.readAllLines(Path)?

A. String
B. String[]
C. List<String>
D. Stream<String>
Correct Answer: C. List<String>

Q9. What best prevents a FileInputStream resource leak?

A. Call System.gc()
B. Use try-with-resources
C. No fix is needed
D. Only wrap read() in try-catch
Correct Answer: B. Use try-with-resources

Q10. A serializable Employee extends non-serializable Person. What must Person provide for deserialization?

A. serialVersionUID
B. No-argument constructor
C. transient fields
D. clone() method
Correct Answer: B. No-argument constructor

Q11. Which stream pair copies a JPG image without data loss?

A. FileReader/FileWriter
B. BufferedReader/BufferedWriter
C. FileInputStream/FileOutputStream
D. InputStreamReader/OutputStreamWriter
Correct Answer: C. FileInputStream/FileOutputStream

Q12. Opening a missing report.txt using FileReader throws:

A. IOException
B. FileNotFoundException
C. NullPointerException
D. NoSuchFileException
Correct Answer: B. FileNotFoundException

Q13. What does BufferedWriter.newLine() do?

A. No difference from write newline
B. Writes platform-specific line separator
C. Runs faster than write
D. Automatically flushes
Correct Answer: B. Writes platform-specific line separator

Q14. NotSerializableException most likely occurs because:

A. serialVersionUID is missing
B. A field is non-serializable and not transient
C. Class is final
D. No explicit constructor
Correct Answer: B. A field is non-serializable and not transient

Q15. Scanner with delimiter comma over "10,20,30" prints:

A. 10 20 30
B. 10,20,30
C. Compilation error
D. Infinite loop
Correct Answer: A. 10 20 30

Q16. Without try-with-resources, which method ensures buffered data reaches the file?

A. write()
B. flush() or close()
C. read()
D. available()
Correct Answer: B. flush() or close()

Q17. If directory temp does not exist, two consecutive mkdir() calls return:

A. true, true
B. true, false
C. false, false
D. Compilation error
Correct Answer: B. true, false

Q18. Which class treats a String as a character input stream?

A. StringBuffer
B. StringReader
C. CharArrayWriter
D. StringTokenizer
Correct Answer: B. StringReader

Q19. In new PrintWriter(new FileWriter("out.txt"), true), true enables:

A. Append mode
B. Auto-flush after println()
C. UTF-8 encoding
D. Internal buffering
Correct Answer: B. Auto-flush after println()

Q20. Files.copy(source,target,REPLACE_EXISTING) when target exists:

A. Throws FileAlreadyExistsException
B. Skips copy
C. Overwrites target
D. Appends source
Correct Answer: C. Overwrites target